Governing Equations of the **fuzzy-couscous** Solver

Clean-room compressible LES solver for blast-induced turbulence

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Abstract

This document collects every governing equation the **fuzzy-couscous** solver discretizes: the compressible Navier–Stokes system, the inviscid and viscous fluxes, the three equations of state (ideal gas, two-γ mixture, and Jones–Wilkins–Lee), the multifluid marker transport, the LES closure (hyperdissipation, localized artificial diffusivity, Ducros sensor), the Besnard–Harlow–Rauenzahn (BHR) variable-density turbulence model, the SSP-RK3 time integrator, and the diagnostic identities. Each section cites the file that implements it, so the equations stay traceable to the code.

Contents

1 Compressible Navier–Stokes system

The solver advances the conservative compressible Navier–Stokes equations for a single fluid (or a mixture; see §4) on a periodic / slip-wall Cartesian box. With conserved state $\mathbf{U} = (\rho,\ \rho\mathbf{u},\ \rho E)^{\mathsf{T}}$,

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + p\,\mathbf{I}) = \nabla\cdot\boldsymbol{\tau}, \tag{2}$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla\cdot[(\rho E + p)\,\mathbf{u}] = \nabla\cdot(\boldsymbol{\tau}\cdot\mathbf{u}) - \nabla\cdot\mathbf{q}, \tag{3}$$

where ρ is density, $\mathbf{u} = (u, v, w)$ velocity, p pressure, and the total specific energy is $E = e + \frac{1}{2}|\mathbf{u}|^2$ with e the specific internal energy. The right-hand side is assembled as a telescoping difference of face fluxes, $R_i \;-\!= (F_{i+1/2} - F_{i-1/2})/\Delta x$, so the discrete scheme is conservative by construction (`numerics/RHS.cpp`).

2 Inviscid flux

The hyperbolic (Euler) flux in direction $d \in \{x, y, z\}$ with unit normal $\mathbf{n}_d$ and normal velocity $u_d = \mathbf{u}\cdot\mathbf{n}_d$ is

$$\mathbf{F}_d^{\mathrm{euler}}(\mathbf{U}, p) = \begin{pmatrix} \rho u_d \\ \rho u\,u_d + p\,n_{d,x} \\ \rho v\,u_d + p\,n_{d,y} \\ \rho w\,u_d + p\,n_{d,z} \\ (\rho E + p)\,u_d \end{pmatrix}. \tag{4}$$

It is evaluated by a hybrid 6th-order central / 5th-order WENO reconstruction in flux-difference form with Lax–Friedrichs splitting; the WENO weight is gated by the shock sensor of §6 (`physics/EulerFlux.hpp`, `numerics/WENO5.hpp`). The flux is EOS-agnostic: p is supplied by §4.

3 Viscous fluxes

The viscous stress is the compressible Stokes tensor with an optional bulk viscosity μ_b,

$$\tau_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \tfrac{2}{3}\,\delta_{ij}\,\nabla\cdot\mathbf{u}\right) + \mu_b\,\delta_{ij}\,\nabla\cdot\mathbf{u}, \tag{5}$$

and the heat flux is Fourier's law

$$\mathbf{q} = -\kappa\,\nabla T, \qquad \kappa = \frac{\mu\,c_p}{\mathrm{Pr}}, \tag{6}$$

with T the temperature and Pr the Prandtl number. First derivatives use composed 6th-order central stencils (NGHOST $= 6$), so the viscous Laplacian is 6th-order accurate (`physics/ViscousFlux.hpp`, `numerics/Gradients.cpp`).

4 Equation of state

The pressure closure maps $(\text{marker}, \rho, e) \mapsto (p, c)$ through a single mixture dispatcher (`physics/MixtureEOS.hpp`), where c is the sound speed used at the flux and CFL sites.

4.1 Ideal gas

$$p = (\gamma - 1)\,\rho e, \qquad c^2 = \frac{\gamma p}{\rho}. \tag{7}$$

4.2 Two-γ mixture (variable-density contact)

A per-cell advected marker $G = 1/(\gamma - 1)$ selects between ambient air and a second ideal gas (products with γ_p). The effective ratio of specific heats is recovered as $\gamma = 1 + 1/G$, giving

$$p = \frac{\rho e}{G}, \qquad c^2 = \frac{(1 + 1/G)\,p}{\rho}. \tag{8}$$

This is the original variable-density contact model (`physics/Multifluid.hpp`).

4.3 Jones–Wilkins–Lee (JWL) for real high explosives

Detonation products (e.g. TNT) obey the JWL EOS, which an ideal gas cannot represent. With relative volume $V = \rho_0/\rho$ (ρ_0 the reference/solid density) and Grüneisen coefficient ω,

$$p(\rho, e) = A\left(1 - \frac{\omega}{R_1 V}\right) e^{-R_1 V} + B\left(1 - \frac{\omega}{R_2 V}\right) e^{-R_2 V} + \omega\,\rho\,e. \tag{9}$$

Writing $f(V)$ for the reference-curve (first two) terms, $p = f(V) + \omega\rho e$, which inverts in closed form for the internal energy at fixed density, $e = (p - f(V))/(\omega\rho)$, and yields the isentropic sound speed $c^2 = (\partial p/\partial\rho)_s$. With $A = B = 0$ the EOS reduces to an ideal gas with $\gamma = 1 + \omega$, the hook that lets the mixture dispatcher treat air and products uniformly (`physics/JWL.hpp`).

The shipped TNT parameter set (LLNL / Dobratz & Crawford) is:

Parameter	Symbol	Value
Reference coefficient	A	371.2 GPa
Reference coefficient	B	3.231 GPa
Reference exponent	R_1	4.15
Reference exponent	R_2	0.95
Grüneisen coefficient	ω	0.30
Reference density	ρ_0	1630 $\mathrm{kg/m^3}$
Detonation energy	E_0	7.0 GPa
CJ pressure	p_{CJ}	21 GPa
Detonation speed	D	6930 m/s

Runs are nondimensionalized with $\rho_{\mathrm{ref}} = \rho_0$ and $p_{\mathrm{ref}} = p_{\mathrm{CJ}}$ so the $\sim$1000:1 density and $\sim 10^5$:1 pressure contrasts (Atwood ≈ 0.999) stay well conditioned. The CJ density $\rho_{\mathrm{CJ}} = \rho_0/(1 - p_{\mathrm{CJ}}/(\rho_0 D^2))$ is the Rankine–Hugoniot value, self-consistent with the JWL sound speed at CJ ($c_{\mathrm{CJ}} = D - u_{\mathrm{CJ}}$).

5 Multifluid marker transport

The marker (the two-γ field G, or the products mass fraction $\phi \in [0, 1]$ for JWL) is advected with the flow,

$$\frac{\partial(\rho\,\chi)}{\partial t} + \nabla\cdot(\rho\,\chi\,\mathbf{u}) = 0, \qquad \chi \in \{G,\ \phi\}. \tag{10}$$

The contact is handled either by a gated double-flux (non-oscillatory) or an opt-in fully conservative flux (`conservative = true`); the latter conserves total energy at the cost of small contact oscillations (`physics/Multifluid.cpp`).

6 LES closure and shock capturing

6.1 Spectral hyperdissipation (LES sink)

The sub-grid sink is a hyper-viscous term applied to every conserved variable,

$$\left.\frac{\partial \mathbf{U}}{\partial t}\right|_{\text{LES}} = -\,\nu_h\,\nabla^4\mathbf{U}, \tag{11}$$

whose discrete operator matches the analytic k^4 damping; under SSP-RK3 it preserves the $e^{-\nu_h k^4 t}$ eigenvalue decay (`numerics/HyperdissipationSpectral.cpp`). The empirical stability invariant is fixed $\nu_{\text{eff}} = \nu_h k_{\max}^2$; between resolutions ν_h scales as Δx^2 (see the README TGV note).

6.2 Ducros shock sensor

The hybrid central/WENO blend and the artificial diffusivity are gated by the Ducros sensor

$$\theta = \frac{(\nabla\cdot\mathbf{u})^2}{(\nabla\cdot\mathbf{u})^2 + |\boldsymbol{\omega}|^2 + \varepsilon}, \tag{12}$$

with vorticity $\boldsymbol{\omega} = \nabla\times\mathbf{u}$ and a small ε; $\theta\to 1$ in compression (shocks), $\theta \to 0$ in smooth/vortical regions. A pressure-jump tanh ramp and a half-width-2 dilation along the face direction sharpen the gate (`numerics/Ducros.cpp`).

6.3 Localized artificial diffusivity (LAD)

For strong blasts the Cook/Kawai–Lele localized artificial diffusivity adds grid-dependent, sensor-localized dissipation to shocks and contacts,

$$\mu^* = C_\mu\,\rho\,|\nabla^r(\nabla\cdot\mathbf{u})|\,\Delta^{r+2}, \qquad \beta^* = C_\beta\,\rho\,|\nabla^r(\nabla\cdot\mathbf{u})|\,\Delta^{r+2}, \tag{13}$$

with an analogous artificial mass/contact diffusivity D^* for the density-contact piece; the canonical order is $r = 4$ (`numerics/ArtificialDiffusivity.cpp`).

7 BHR variable-density turbulence model

The Besnard–Harlow–Rauenzahn k–ε–a–b model runs as an operator-split sub-step alongside the LES hydro. The six primitives are the turbulent kinetic energy k, its dissipation ε, the turbulent mass flux a_i, and the density–specific-volume covariance $b = -\,\overline{\rho'\,(1/\rho)'}$. The dominant TKE source is the variable-density (baroclinic) production

$$P_{\text{VD}} = -\,a_i\,\frac{\partial p}{\partial x_i}, \tag{14}$$

validated as dominant in the LES a/b budget. The turbulent viscosity is $\nu_t = C_\mu\,k^2/\varepsilon$, and the modeled transport equations carry gradient-diffusion terms with Schmidt numbers $(\sigma_k, \sigma_\varepsilon, \sigma_a, \sigma_b)$ (`turbulence/BHR.hpp`, `turbulence/BHR.cpp`).

When two-way feedback is enabled, the model returns a Reynolds-stress divergence and dissipative heating to $\mathbf{U}$, blended by the resolution-adequacy sensor

$$f_k = \frac{k_{\text{sgs}}}{k_{\text{res}} + k_{\text{sgs}}}, \qquad k_{\text{res}} = \tfrac{1}{2}\left|\mathbf{u} - \widetilde{\mathbf{u}}\right|^2, \tag{15}$$

where $\widetilde{\mathbf{u}}$ is a test-filtered velocity (resolved energy in the $[\Delta x, 2\Delta x]$ octave); $f_k \to 1$ under-resolved (RANS), $f_k \to 0$ when the LES resolves the energy.

8 Time integration

Time advance uses the three-stage strong-stability-preserving Runge–Kutta (SSP-RK3, Gottlieb–Shu) scheme on $\mathbf{U}^n \to \mathbf{U}^{n+1}$ with right-hand side $\mathcal{L}(\mathbf{U})$:

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t\, \mathcal{L}(\mathbf{U}^n), \tag{16}$$

$$\mathbf{U}^{(2)} = \tfrac{3}{4}\,\mathbf{U}^n + \tfrac{1}{4}\,\mathbf{U}^{(1)} + \tfrac{1}{4}\,\Delta t\, \mathcal{L}(\mathbf{U}^{(1)}), \tag{17}$$

$$\mathbf{U}^{n+1} = \tfrac{1}{3}\,\mathbf{U}^n + \tfrac{2}{3}\,\mathbf{U}^{(2)} + \tfrac{2}{3}\,\Delta t\, \mathcal{L}(\mathbf{U}^{(2)}). \tag{18}$$

The time step obeys the minimum of a hyperbolic CFL ($\propto \Delta x/(|\mathbf{u}| + c)$) and a viscous limit; in MPI runs both are reduced with `MPI_Allreduce(MIN)` (`numerics/RK3.cpp`, `numerics/RHS.cpp`).

9 Diagnostic identities

9.1 Helmholtz decomposition

The velocity is split into solenoidal and dilatational parts, $\mathbf{u} = \mathbf{u}_s + \mathbf{u}_d$ with $\nabla\cdot\mathbf{u}_s = 0$ and $\nabla\times\mathbf{u}_d = \mathbf{0}$, giving the kinetic-energy split $K = K_{\text{sol}} + K_{\text{dil}}$ and the per-shell spectra $E(k) = E_{\text{sol}}(k) + E_{\text{dil}}(k)$ (`diagnostics/Spectra.cpp`).

9.2 Dissipation budget

The resolved dissipation is split into solenoidal and dilatational contributions,

$$\varepsilon_{\text{total}} = \frac{\mu}{\rho}\left\langle \tau_{ij}\,\partial_j u_i \right\rangle, \qquad \varepsilon_{\text{sol}} = \nu\,\langle|\boldsymbol{\omega}|^2\rangle, \qquad \varepsilon_{\text{dil}} = \tfrac{4}{3}\,\nu\,\langle(\nabla\cdot\mathbf{u})^2\rangle, \tag{19}$$

together with the turbulent Mach number $M_t = u_{\text{rms}}/\langle c\rangle$ (`diagnostics/Statistics.cpp`).

9.3 Conservation monitors

For a closed chamber the totals E/E_0 and M/M_0 are conserved to round-off by the discrete divergence theorem (telescoping face fluxes plus $\mathbf{u}\cdot\mathbf{n} = 0$ at slip walls); the internal energy $e_{\text{int}} = \sum(\rho E - \frac{1}{2}\rho|\mathbf{u}|^2)$ is accumulated directly so $KE + e_{\text{int}} = e_{\text{total}}$ is a genuine consistency check rather than an identity (`diagnostics/Statistics.cpp`; see the README for the conservation-vs-truncation discussion).